

E-Commerce Customer Churn and Spending Prediction Using Machine Learning

Small Project Report / Knowledge Base — Current Methodology

1. Abstract

This project investigates the use of machine learning to analyse customer behaviour in e-commerce and produce two related predictions: (1) whether a customer is likely to churn and (2) the customer's expected future spending. The methodology combines transaction-level e-commerce data from the UCI Online Retail II dataset and the Brazilian E-Commerce Public Dataset by Olist. The raw sources have different schemas and business contexts, so the project focuses on transforming transactional information into a customer-level spending/churn modelling dataset. Random Forest is used as the main classification model for churn prediction and Random Forest Regressor is used for customer spending prediction. The spending target currently being implemented is the customer's next-order spending target. The overall workflow covers data preparation, customer-level feature engineering, target construction, train/test splitting, model training and evaluation.

2. Introduction / Project Overview

Customer retention and customer value are two closely related problems in e-commerce analytics. A business needs to understand which customers may stop purchasing and, at the same time, estimate the value of future purchases. Transaction data provides behavioural signals such as purchase frequency, recency, quantity and monetary value that can be converted into machine-learning features.

The project is intentionally designed as a practical, manageable machine-learning research project rather than an overly complex predictive system. The objective is to establish a clear end-to-end methodology that can be implemented, evaluated and documented using standard supervised-learning techniques.

3. Core Problem

E-commerce transaction tables are recorded at the transaction or order-item level, while the business questions of churn and future spending are customer-level questions. Directly applying a model to raw rows would therefore mix different levels of observation and could produce misleading results. The core technical problem is to convert historical transactions into a customer-level modelling table while keeping the temporal relationship between past behaviour and future outcomes intact.

4. Research Question

Main research question: Can customer-level behavioural features derived from historical e-commerce transactions be used with standard machine-learning models to predict customer churn and future spending?

Supporting questions: (a) Which historical purchasing characteristics can be transformed into useful customer-level predictors? (b) Can a Random Forest classifier identify customers associated with the churn target? (c) Can a Random Forest Regressor estimate a customer's next-order spending target?

5. Datasets Used

Dataset	Source / Role	Important characteristics
UCI Online Retail II	UCI Machine Learning Repository	Two years of transaction data from a UK-based registered non-store online retailer; includes
Brazilian E-Commerce Public Dataset	Knowledge Base by Olist	About 100,000 orders from 2016–2018 in Brazil, with linked customer, order, item, payment

The project work has been moving toward a single customer/spending modelling dataset rather than training directly on the raw source tables. The combined dataset is intended to provide a consistent modelling structure with approximately 100,000 records for the spending-prediction workflow. The exact final row count and column list

should be taken from the final generated CSV/notebook output when the implementation is frozen.

6. Proposed Solution / Methodology

Step 1 — Data acquisition: Obtain the UCI Online Retail II and Olist Brazilian E-Commerce datasets.

Step 2 — Data cleaning: Handle missing customer identifiers, invalid or cancelled transactions where appropriate, date parsing, duplicate/quality issues, and inconsistent data types.

Step 3 — Data integration: Join the relevant Olist tables through their keys and transform the source data into a customer-oriented analytical structure.

Step 4 — Temporal feature engineering: Use historical transaction behaviour to construct customer-level predictors. Typical behavioural variables include recency, purchase/order frequency, quantity and monetary/spending measures, subject to the final implementation.

Step 5 — Target construction: Create the churn target from subsequent customer activity and construct the future-spending target. The spending implementation currently uses the next-order spending target.

Step 6 — Prepare X and y: Separate predictor variables (X) from the target variable (y), while excluding identifiers or target-derived information that should not be used as predictors.

Step 7 — Train/test split: Split the modelling data into training and testing portions before fitting the final models.

Step 8 — Model training: Train a Random Forest Classifier for churn prediction and a Random Forest Regressor for customer spending prediction.

Step 9 — Evaluation: Evaluate classification and regression performance with metrics appropriate to each task and inspect model behaviour before drawing conclusions.

7. Models Used

Task	Model	Output
Customer churn prediction	Random Forest Classifier	Classification of the churn target.
Customer spending prediction	Random Forest Regressor	Prediction of the customer's next-order spending amount.

8. Feature / Target Design

The important methodological distinction is between **historical predictors** and **future targets**. Features must be calculated using information available before the prediction point. For spending prediction, the current implementation uses the next-order spending amount as the target, so the model learns a relationship between prior customer behaviour and the monetary value of the next purchase. For churn, the target is based on subsequent customer activity rather than simply labelling a customer from the same transactions used to create the predictors. This temporal separation is important to reduce target leakage.

The final implementation should preserve a clear observation period and prediction period. Dates, cut-off rules, and any inactivity threshold used for the churn label should be recorded in the final methodology so the experiment is reproducible.

9. Current Implementation Status

Component	Current status
Dataset selection	UCI Online Retail II + Olist Brazilian E-Commerce selected.
Combined modelling data	Being prepared as a unified spending/customer dataset; approximately 100,000 records are targeted for th
Feature engineering	Customer-level historical behavioural features are being prepared.
Spending target	Next-order spending target selected and used for the regression workflow.

Component	Current status
X and y preparation	Completed/under implementation as part of Task 6.1 workflow.
Train/test split	Task 6.2 workflow established.
Random Forest Regressor	Task 6.3 workflow established.
Churn classifier	Random Forest Classifier is the planned classification model.

10. Expected Evaluation

For the churn classifier, evaluation should focus on classification metrics such as accuracy, precision, recall, F1-score and a confusion matrix, with particular attention to class imbalance if it exists. For the spending regressor, suitable measures include MAE, RMSE and R^2 . These metrics describe model performance; they should be reported from the held-out test data and not from the training data alone.

11. Scope and Limitations

- The two source datasets represent different e-commerce businesses, countries and time periods; they should not be treated as one naturally occurring commercial dataset.
- The project is primarily a methodology and predictive-modelling exercise. It does not claim that the trained models represent current real-world customer behaviour.
- Customer identifiers and source-specific fields should be handled carefully during integration to avoid artificial relationships or leakage.
- The exact churn definition and threshold must be explicitly documented in the final implementation because churn is not a directly supplied target in the raw transaction data.
- The final conclusions should be based on the actual held-out evaluation results rather than assumptions about model performance.

12. Project Objective in One Statement

The project aims to build a straightforward, reproducible machine-learning pipeline that converts historical e-commerce transactions into customer-level behavioural features and uses Random Forest models to predict customer churn and next-order spending.

13. Source References

1. UCI Machine Learning Repository — Online Retail II, Dataset 502. The dataset contains 1,067,371 transaction records from a UK-based non-store online retailer covering 01/12/2009–09/12/2011. <https://archive.ics.uci.edu/dataset/502/online+retail+ii>
2. Kaggle / Olist — Brazilian E-Commerce Public Dataset. The dataset contains information for about 100,000 orders from 2016–2018, with linked customer, order, item, payment, review, product and seller information. <https://www.kaggle.com/olistbr/brazilian-ecommerce>

Prepared as a project knowledge-base document. This report records the methodology currently established in the project; final numerical results, exact feature names, hyperparameters and final churn-threshold definition should be added after the implementation is completed.